

CHAPTER 3

Activities & Lab Packet

Water on Earth

A complete set of labs, worksheets, writing prompts, discussions, and group projects to accompany Chapter 3. Works with both the Standard and Easy Read versions of the textbook.

 Georgia Standard S6E3

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1 Water Cycle in a Bag

LAB

OBJECTIVE

Build a mini water cycle and observe evaporation, condensation, and "precipitation" powered by the Sun.

TIME

20 min setup + observations over 2–3 days

STANDARDS

S6E3.b

Materials (per group of 2–3)

- 1 zip-top sandwich bag (sturdy, no leaks)
- About 60 mL ($\frac{1}{4}$ cup) of water + 2 drops of blue food coloring (it's the "ocean")
- Permanent marker, strong tape
- A sunny window; student data sheet (next page)

Setup

Do this on a sunny stretch of days if possible. Have each group draw a sun, a cloud, and ocean waves on their bag with marker BEFORE adding water — it turns the bag into a labeled diagram of itself.

Student Instructions

1. Draw a sun in one top corner, a cloud in the other, and waves along the bottom of your bag.
2. Pour in the blue water, squeeze out most of the air, and seal the bag completely.
3. Tape the bag to a sunny window with the water at the bottom. Record the date, time, and what the bag looks like (Day 0 sketch).
4. Check the bag each day for 2–3 days. Look closely near the top and sides: droplets? fog? drips sliding down? Sketch and describe what you see each day.
5. Identify where in YOUR bag each step happens: evaporation, condensation, precipitation, collection. Complete the data sheet.

⚠ SAFETY & TEACHER NOTES

Tip: Double-check seals — a leaky bag on a windowsill is a sad Tuesday. Tape all four corners.

Key connector question: "What's the energy source heating your mini ocean? Could the cycle run in a dark closet?" (Try one bag in a closet as a class control — great experimental design talk!)

Common confusion: Students may say the water "disappeared." Push for the word *evaporated* — it became invisible water VAPOR, still inside the bag.

ANSWER KEY (data sheet)

1) Droplets/fog form near the top and slide down the sides | 2) Evaporation = water warmed at the bottom becomes vapor; condensation = droplets forming on the cool plastic; precipitation = drops growing heavy and sliding/falling; collection = water pooling at the bottom again | 3) The Sun — without it (closet bag) almost nothing happens | 4) No water escaped; the same water keeps cycling — just like Earth's closed water cycle.

Water Cycle in a Bag — Data Sheet

Name: _____

Period: _____

Day	Sketch of the bag	What I observed (droplets? fog? drips?)
Day 0		
Day 1		
Day 2–3		

1. Where did you first notice water droplets forming? Why there?

2. Match each water-cycle step to where it happened in YOUR bag:

Step	Where I saw it in the bag
Evaporation	
Condensation	
Precipitation	
Collection	

3. What is the energy source driving your mini water cycle? What's your evidence?

Conclusion: The bag stayed sealed the whole time. What does that tell you about Earth's water supply?

2 Sonar Shoebox — Mapping a Hidden Ocean Floor

LAB

OBJECTIVE

Map an unseen "ocean floor" with depth probes, the way scientists map the real one with sonar.

TIME

40–50 minutes

STANDARDS

S6E3.c

Materials (per group of 3–4)

- 1 shoebox with a lid; modeling clay or crumpled foil INSIDE forming a secret landscape (built by the teacher or a rival group): include a tall ridge, a deep trench, a flat plain, and a shelf near one end
- Lid with a straight row of 10–12 small holes punched along the center line, numbered
- 1 wooden skewer marked in centimeters (the "sonar ping")
- Graph paper and the student data sheet (next page)

Setup

Build (or have groups secretly build and swap) clay landscapes in the boxes, then tape the lids shut. Groups must NEVER open the box — that's the whole point. Real oceanographers can't drain the ocean to peek either!

Student Instructions

1. Lower the skewer gently through hole #1 until it touches bottom. Read the depth (cm from the lid) and record it in your data table.
2. Repeat for every hole, in order. No peeking, no tilting the skewer.
3. On graph paper, plot depth (y-axis, increasing DOWNWARD) against hole number (x-axis). Connect the dots — that's your depth profile, just like a sonar trace!
4. Label what you think each part of your profile is: shelf, slope, abyssal plain, ridge, trench.
5. Finally, open the box and compare your map to reality. How close was your profile?

⚠ SAFETY & TEACHER NOTES

Safety: Skewer points are pokey — probe slowly, hands clear of holes.

The big idea: Sonar measures the TIME for sound to bounce back; our skewer measures distance directly. Both build a picture without seeing.

Discussion starter at end: "Your profile missed features between the holes. How could you make a better map?" (More holes = more sonar pings = higher resolution!)

ANSWER KEY (data sheet)

Profiles vary by box | 3) Deepest reading = trench; tallest bump = mid-ocean ridge; long even stretch = abyssal plain; shallow zone near the end = continental shelf | 4) More holes / closer spacing | Conclusion: scientists use sonar (sound pings) and satellites; depth-by-depth measurements get combined into a map of terrain no one has ever seen directly.

Sonar Shoebox — Data Sheet

Name: _____

Period: _____

Depth readings:

Hole #	1	2	3	4	5	6	7	8	9	10	11	12
Depth (cm)												

1. Attach your graph-paper profile. Remember: depth increases DOWNWARD, hole numbers go left to right.

2. Label your profile. Write which hole numbers you believe cover each feature:

Feature	Hole numbers	My evidence (what the depths did)
Continental shelf		
Continental slope		
Abyssal plain		
Mid-ocean ridge		
Trench		

3. After opening the box: what did your map get right? What did it miss?

4. How could you change the lid to make a more accurate map?

Conclusion: How is your skewer like real sonar? How is it different?

3 Earth's Water in a Bucket — Scale It Down

LAB

OBJECTIVE

Measure out Earth's water distribution to scale and discover how little usable fresh water exists.

TIME

30 minutes

STANDARDS

S6E3.a

Materials (per group of 3–4)

- 1,000 mL of water in a large beaker or pitcher — this represents ALL of Earth's water
- Graduated cylinders (100 mL and 10 mL), an eyedropper
- 4 labeled cups: **OCEANS, ICE & GLACIERS, GROUNDWATER, RIVERS & LAKES**
- Blue food coloring (optional, for drama); student data sheet (next page)

Setup

Before measuring, have each group **PREDICT** by drawing lines on their cups: how much of the 1,000 mL do they think goes in each? Most students wildly overestimate rivers and lakes — let the data do the correcting.

Student Instructions

1. Record your group's predictions for each cup on the data sheet.
2. Measure **970 mL** into the OCEANS cup. That's the 97% salt water.
3. From the remaining 30 mL (all the fresh water on Earth!), measure about **20 mL** into ICE & GLACIERS — frozen and locked away.
4. Measure about **9 mL** into GROUNDWATER — fresh, but hidden underground.
5. Use the eyedropper to move the last **~1 mL** (a few drops!) into RIVERS & LAKES. That tiny splash is the surface fresh water humans use most easily.
6. Line up the four cups, compare to your predictions, and complete the data sheet.

SAFETY & TEACHER NOTES

Tip: Water spills — towels nearby. Coloring the final 1 mL blue makes the "this is all we get" moment land harder.

Discussion starter at end: "The ice cup holds most of the fresh water. What happens to sea levels — and to the fresh-water supply — if it melts into the ocean cup?"

Connection: Ties directly to Chapter 6 conservation — keep the cups for that unit!

ANSWER KEY (data sheet)

Measured: oceans 970 mL, ice ~20 mL, groundwater ~9 mL, rivers/lakes ~1 mL | 2) Salt water — 97% | 3) Ice & glaciers | 4) About 1 mL out of 1,000 mL — about 0.1%, less than 1% | Conclusion: usable fresh water is extremely rare, so protecting rivers, lakes, and groundwater from pollution and waste is critical.

Earth's Water in a Bucket — Data Sheet

Name: _____

Period: _____

Cup	My PREDICTION (mL)	Actual measured (mL)	Surprised? Why?
Oceans (salt)			
Ice & glaciers			
Groundwater			
Rivers & lakes			

1. Which cup were you most wrong about in your prediction? By how much?

2. Most of Earth's water is (circle one): SALT WATER / FRESH WATER. What percent? _____

3. Where is MOST of Earth's FRESH water stored?

4. How many mL out of 1,000 ended up in "Rivers & lakes"? About what percent of all water is that?

5. The textbook says less than 1% of Earth's water is fresh water people can easily use. Does your model agree? Explain.

Conclusion: Based on this lab, why is protecting fresh water so important? Give one specific way to protect it.

4 Where's the Water?

WORKSHEET

OBJECTIVE

Identify where Earth's water is located, and how salt and fresh water are distributed.

TIME

15 minutes

STANDARDS

S6E3.a

Name: _____

Period: _____

Part A. Fill in the numbers (71, 97, 3, 1):

1. About _____% of Earth's surface is covered by water.
2. About _____% of Earth's water is salty ocean water.
3. Only about _____% of Earth's water is fresh.
4. Less than _____% of all water is fresh water people can easily reach and use.

Part B. Match each term to its description (write the letter):

Term	Descriptions
____ 5. Groundwater	A. An underground layer of rock or soil that stores water; wells pump from it B. Fresh water stored underground in soil and rock C. Frozen fresh water at the poles and in mountains — most of Earth's fresh water D. The small share of surface fresh water that's easiest for people to use
____ 6. Ice caps & glaciers	
____ 7. Aquifer	
____ 8. Rivers & lakes	

Part C. Short answers:

9. Earth has SO much water. Why can't we just use ocean water for drinking and farming?

10. A town gets its drinking water from wells. Where is that water actually coming from, and what is the storage layer called?

11. Explain why "Earth is covered in water" and "fresh water is rare" can BOTH be true.

ANSWER KEY

1) 71 2) 97 3) 3 4) 1 | 5) B 6) C 7) A 8) D | 9) It's too salty to drink or water crops | 10) Groundwater pumped from an aquifer | 11) Most water (97%) is salty, and most fresh water is frozen — so the easily usable slice is tiny even on a watery planet.

5 Water Cycle Diagram & Steps

WORKSHEET

OBJECTIVE

Name, order, and explain the steps of the water cycle and its energy source.

TIME

20 minutes

STANDARDS

S6E3.b

Name: _____

Period: _____

Part A. Fill in each blank with: *evaporation, condensation, precipitation, runoff/collection, transpiration*:

1. The Sun heats the ocean and liquid water becomes rising vapor: _____
2. High up, cooling vapor turns into droplets and clouds form: _____
3. Heavy drops fall as rain, snow, sleet, or hail: _____
4. Water flows over land into rivers, lakes, and oceans: _____
5. Plants release water vapor from their leaves: _____

Part B. Complete the table:

Step	Water changes from → to	Driven by (Sun's heat, cooling, or gravity?)
Evaporation		
Condensation		
Precipitation		

Part C. Short answers:

6. What would happen to the water cycle if the Sun "turned off"? Which step would stop FIRST?

7. A puddle disappears on a hot afternoon. Where did the water go? (Be scientific — it didn't vanish!)

8. The Sun lifts water up. What force brings it back down? Name TWO steps that force drives.

ANSWER KEY

1) evaporation 2) condensation 3) precipitation 4) runoff/collection 5) transpiration | B: evaporation = liquid → gas (Sun's heat); condensation = gas → liquid droplets (cooling); precipitation = droplets → falling rain/snow (gravity) | 6) Evaporation stops first — no heat to lift water; the whole cycle shuts down | 7) It evaporated — became invisible water vapor in the air | 8) Gravity; precipitation and runoff/collection.

6 Wave, Current, or Tide?

WORKSHEET

OBJECTIVE

Distinguish waves, currents, and tides by what moves and what causes each.

TIME

15 minutes

STANDARDS

S6E3.d

Name: _____

Period: _____

Part A. Write **W** (wave), **C** (current), or **T** (tide):

- | | |
|---|---|
| _____ 1. Energy moves; the water mostly bobs in place | _____ 6. Deep ones are driven by temperature & saltness |
| _____ 2. A steady river of water flowing in the ocean | _____ 7. Moves heat around the planet, shaping climate |
| _____ 3. The daily rise and fall of the sea | _____ 8. The beach is wide at noon, underwater by |
| _____ 4. Caused mostly by the Moon's gravity | evening |
| _____ 5. Caused mostly by wind on the surface | |

Part B. Complete the comparison table:

	WAVE	CURRENT
What moves?		
Main cause?		

Part C. Short answers:

9. Explain the "stadium wave" comparison: how is a crowd wave like an ocean wave?

10. About how many high tides does a typical beach get each day, and why? (Hint: bulges + Earth's rotation.)

11. When the Sun and Moon line up, what happens to the tides, and why?

ANSWER KEY

1) W 2) C 3) T 4) T 5) W 6) C 7) C 8) T | B: wave — energy moves / wind; current — the water itself moves in one direction / wind plus temperature & saltness | 9) People stand and sit but don't run around the stadium; energy travels while the "medium" stays put | 10) Two — Earth rotates through two tidal bulges each day | 11) Extra-large tides — their gravitational pulls combine.

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain concepts in writing.

TIME

15–30 minutes each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Diary of a Water Drop.** You are a drop of ocean water. Tell the story of one trip through the water cycle — name each step as it happens to you.
2. **The Blue Marble.** Astronauts say Earth looks blue from space. Explain why — and then explain the surprising catch about how little of that water we can actually drink.
3. **Beach Day Mystery.** You built a sandcastle near the water at 10 a.m. By 4 p.m. it's underwater — but there was no storm. Explain to a confused friend what happened.

Level 2 — Intermediate (3 paragraphs)

4. **The Sun Goes on Strike.** Imagine the Sun stopped shining for one year (Earth somehow stays warm enough to live). Describe what would happen to the water cycle, step by step, and what that would mean for rain and rivers.
5. **Hidden Mountains.** The tallest mountain ranges and deepest valleys on Earth are underwater. Describe a submarine journey from a beach to the bottom of a trench, naming each ocean-floor feature you pass in order.
6. **Wave vs. Current: The Debate.** A wave and a current argue about which is more important to Earth. Write their debate — each must correctly describe what it moves, what causes it, and one big effect it has.

Level 3 — Advanced (essay)

7. **Water Budget Argument.** Some say "Earth will never run out of water, so we don't need to conserve it." Using the salt/fresh percentages and the water cycle, write an argument explaining what's right AND wrong about that claim.
8. **Mapping the Unseen.** We have better maps of the Moon's surface than of our own ocean floor. Explain why mapping the ocean floor is so hard, how sonar works, and argue whether ocean exploration deserves more funding.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific thinking through verbal reasoning.

USE

Warm-ups, exit tickets, Socratic seminar

STANDARDS

Cross-cutting

TEACHER NOTES

These are designed for "think-pair-share" format. Give students 30 seconds to think, 1 minute to discuss with a partner, then call on a few groups. Some questions have no single "right" answer — push students to defend their thinking with evidence from the chapter.

Quick Warm-Ups (1–2 min each)

1. Is the water you drank today "new" water? Where might it have been a million years ago?
2. Why doesn't Earth run out of rain?
3. You see your breath fog up a cold window. Which water-cycle step did you just watch?
4. Which holds more fresh water: all the world's rivers, or the ice at the poles?
5. The tide is coming in. Is the Moon doing something different right now, or is Earth?

Deeper Discussions (5–15 min each)

6. Salt water is 97% of Earth's water. Why don't we just remove the salt? (What might it cost in money and energy — and who could afford it?)
7. Ocean currents move heat around the planet. What might happen to the climate of a coastal city if a major warm current near it slowed down or stopped?
8. If most fresh water is locked in glaciers, what happens to the fresh-water supply as glaciers melt into the salty ocean? Is melting ice "adding" usable water or losing it?
9. Scientists map the ocean floor without ever seeing most of it. What other things in science do we "know" without direct seeing — and what makes that knowledge trustworthy?
10. A drought hits your region. Using the water cycle, brainstorm: where did the rain "go," and why can't we just make more?

9 Journey of a Drop — Comic Strip

GROUP PROJECT

OBJECTIVE

Storyboard one water molecule's journey through the full water cycle with accurate science.

TIME

2 class periods

STANDARDS

S6E3.a, S6E3.b

Project Description

Teams of 2–3 create an 8-panel comic strip following one water drop ("give it a name and a personality!") through a complete trip around the water cycle. The drop must visit at least **4 different locations** on Earth (e.g., ocean → cloud → mountain snow → river → aquifer) and pass through **every step** of the cycle at least once.

Comic Requirements

- **8 panels minimum**, drawn or digital, with dialogue or captions
- Every water-cycle step appears and is **labeled**: evaporation, condensation, precipitation, collection/runoff (bonus: transpiration!)
- The energy source (Sun) and the downward force (gravity) each appear in at least one panel
- At least one panel set in a place from section 3.2: ocean, glacier, groundwater/aquifer, river, or lake
- Science must be accurate — the drop can be funny, but the physics can't be wrong!

Presentation Day

Teams read their comics aloud (2 minutes). The class checks each comic with a "cycle checklist" — did every step appear correctly? Display the strips in a hallway "river" running down the wall.

TEACHER NOTES

Differentiation: Provide a pre-drawn 8-panel template with the first panel started. Advanced teams: require transpiration AND a stop in an aquifer.

Science check: The #1 error is water "disappearing" instead of evaporating, and clouds "being" steam. Catch these in the storyboard stage.

Grading rubric (suggested): Scientific accuracy (40%), all required steps present (25%), creativity (20%), presentation (15%).

10 Vocabulary Bingo

REVIEW GAME

OBJECTIVE

Review chapter vocabulary in a low-stress game format.

TIME

20–30 minutes

STANDARDS

S6E3 (all)

How to Play

1. Students write one word from the word bank into each square of their board, in any order. (FREE space stays free.)
2. The teacher reads a **definition** (not the word!). Students mark the square if they have the matching word.
3. First to 5 in a row calls "BINGO!" — but must read back each marked word AND its meaning to win.
4. Play again with "blackout" rules (whole board) for a longer review.

Word Bank (24 words + FREE)

salt water	evaporation	continental slope	tide
fresh water	condensation	abyssal plain	sonar
groundwater	precipitation	mid-ocean ridge	water vapor
aquifer	runoff	trench	high tide
glacier	transpiration	wave	low tide
water cycle	continental shelf	current	ocean

		FREE 		

TEACHER NOTES

Read definitions straight from the Vocabulary Reference sheet (Section 3) so wording matches what students studied. Keep a checklist of called words to verify BINGOs. For struggling readers, display the word bank on the board during play.